

Tuesday 09/02/2021

Resit Exam

Duration: 90 minutes

Name:

Student No:

P1 [30 points] Decidability

- a) Let $ALLONES_{DFA} = \{\langle D \rangle \mid D \text{ is a DFA that accepts all binary strings that consists of only 1s}\}$. (If $1, 11, 111, 1111, \dots \in L(D)$ then $\langle D \rangle \in ALLONES_{DFA}$) Prove that $ALLONES_{DFA}$ is decidable.
- b) Let $INFINT_{DFA} = \{\langle D_1, D_2 \rangle \mid D_1 \text{ and } D_2 \text{ are DFAs such that } L(D_1) \cap L(D_2) \text{ contains infinitely many strings}\}$. Prove that $INFINT_{DFA}$ is decidable.

P2 [30 points] Undecidability

- a) Let $FINTER_{TM} = \{\langle M_1, M_2 \rangle \mid M_1 \text{ and } M_2 \text{ are TMs such that they accept finitely many common strings}\}$. Prove that $FINTER_{TM}$ is undecidable.
- b) Let $TRIANGLE_{TM} = \{\langle M_1, M_2, M_3 \rangle \mid M_1, M_2 \text{ and } M_3 \text{ are TMs such that } M_1 \text{ accepts } \langle M_2 \rangle, M_2 \text{ accepts } \langle M_3 \rangle, \text{ and } M_3 \text{ accepts } \langle M_1 \rangle\}$. Prove that $TRIANGLE_{TM}$ is undecidable.

P3 [40 points]

- (a) Let $PARTED-SUM = \{\langle B, R, t \rangle \mid B = \{b_1, b_2, \dots, b_m\}, R = \{r_1, r_2, \dots, r_n\} \text{ and for some } \{x_1, x_2, \dots, x_k\} \subseteq B \text{ and } \{y_1, y_2, \dots, y_l\} \subseteq R, \text{ and } k, l \geq 1 \text{ we have } \sum x_i + \sum y_i = t\}$. (In other words, given m blue numbers, n red numbers, and a number t can we find one nonempty subset of blue numbers, one nonempty subset of red numbers so that the total sum of these two subsets is t ?) Prove that $PARTED-SUM$ is NP-complete.
- (b) Let $BLUE-GROUP = \{\langle G, k \rangle \mid G \text{ is a graph with half of the edges are blue, half of the edges are red and it contains a subgraph with } k \text{ vertices that is completely connected with all blue edges (with no red edges)}\}$. Prove that $BLUE-GROUP$ is NP-complete.

- $SAT = \{\langle \phi \rangle \mid \phi \text{ is a satisfiable Boolean formula}\}$. (p.249)
- $3SAT = \{\langle \phi \rangle \mid \phi \text{ is a satisfiable 3cnf Boolean formula}\}$. (p.251)
- $CLIQUE = \{\langle G, k \rangle \mid G \text{ is a undirected graph with a } k\text{-clique}\}$. (clique: complete graph, meaning all vertex pairs have an edge in between. n -clique is a clique with n vertices.)
- $VERTEX-COVER = \{\langle G, k \rangle \mid G \text{ is an undirected graph that has a } k\text{-node vertex cover}\}$. (If G is an undirected graph, a vertex cover of G is a subset of the nodes where every edge of G touches one of those nodes. The vertex cover problem asks whether a graph contains a vertex cover of a specified size.) (p.261)
- $HAMPATH = \{\langle G, s, t \rangle \mid G \text{ is a directed graph with a Hamiltonian path from } s \text{ to } t\}$. (A Hamiltonian path in a directed graph G is a directed path that goes through each node exactly once. $HAMPATH$ problem asks whether there is such a path from a given node to another given node.) (p.242)
- $UHAMPATH$ is the undirected version of $HAMPATH$. (p.267)
- $SUBSET-SUM = \{\langle S, t \rangle \mid S = \{x_1, x_2, \dots, x_k\} \text{ and for some } \{y_1, y_2, \dots, y_l\} \subseteq \{x_1, x_2, \dots, x_k\}, \text{ we have } \sum y_i = t\}$. (p.246)